AI in Banking & Financial Services

An AI strategy for fraud detection, faster credit decisions and personalised banking at a large Indian bank, built around regulatory compliance and data privacy

Executive summary

The bank in this case has three complaints on its desk: fraud cases are going up, loans take too long to approve, and customers do not engage much with the bank beyond the basics. These look like three separate problems owned by three separate teams. I think they share one root: the bank knows a great deal about its customers, but that knowledge sits in different systems, is checked by hand, and reaches decisions late. AI is useful here because it can read signals faster than a person can. It is dangerous here for a different reason, which is that a bank's decisions touch people's money and must be explainable to a customer, an auditor and the Reserve Bank of India (RBI).

My recommendation is a hybrid strategy rather than a single big AI programme. Keep the existing policy rules and regulatory checks. Add machine-learning scores and network analysis for fraud, document extraction plus an explainable risk model for lending, and a grounded multilingual assistant with permission-based recommendations for customer service. Put one shared data and governance layer under all three, so each use case does not rebuild its own data pipeline and its own controls.

The programme runs over 18 months with four review gates. Nothing moves from pilot to scale until the bank has measured a baseline, tested fairness, and shown that customers can reach a human when the system is wrong. The sections that follow explain the diagnosis, the design of each solution, how compliance and privacy are built in, and how success is measured.

1. What is actually going wrong

The case gives no performance figures, so I start from the symptoms and work back to likely causes. I assume a medium-to-large Indian retail and commercial bank with UPI and card channels, a mobile app, retail loans and an existing fraud rule engine. The causes below are hypotheses that the first three months of baseline work should confirm or reject.

Symptom	Likely cause inside the bank	What AI can do	What AI must not do
Rising fraud (UPI scams, card-not-present, account takeover, mule accounts)	Static rules written for yesterday's attacks; signals in separate systems; alerts reviewed late; little view of linked accounts	Score each event in near real time; spot unusual behaviour; map networks of related accounts; rank the alert queue	Block customers silently, or flag so much that genuine payments fail
Slow loan approvals	Paper and PDF documents re-keyed by staff; the same data verified several times; every case queued the same way	Extract fields from documents; verify against source data; route clear cases straight through and borderline cases to officers	Approve on a score alone, or hide the reason for a decline

Symptom	Likely cause inside the bank	What AI can do	What AI must not do
Low customer engagement	Generic campaigns; app used only for balance checks; service in English-heavy channels; no single customer view	Recognise the service need; suggest relevant tools at the right time; answer in the customer's language	Push products the customer does not need, or use data the customer never agreed to share

Two features of the Indian context shape everything that follows. First, payments are instant and irrevocable on rails such as UPI and IMPS, so a fraud decision often has seconds, not hours. Second, the customer base is diverse in language, income pattern and digital comfort. A model or assistant that works for a salaried customer in a metro may fail a trader in a small town, and the bank is accountable for both.

Strategic objective

My objective is to make banking safer, faster and more useful without weakening customer rights or the bank's ability to explain and audit what it did. I would use the RBI FREE-AI Committee report (August 2025) as the benchmark for responsible adoption, and treat the RBI's directions on fraud risk, IT governance and digital lending as binding requirements rather than guidance [1]-[4].

2. Strategic approach and data foundation

Pick the method that fits the decision

One technique will not fit all three problems. Rules suit requirements that must be applied the same way every time, such as sanction checks or product eligibility. Supervised learning suits cases where history provides labels, such as confirmed fraud or past defaults. Anomaly detection helps when labels are scarce or attackers have changed method. Graph analysis finds connections between accounts that no single transaction reveals. Recommendation models rank possible actions for a customer. Generative AI is good with language, so I would use it to answer questions and summarise documents, but I would not let it make the final call on a loan or a fraud block.

The practical rule is to start with the simplest approach that works and move to a more complex model only when testing shows a clear gain that the bank can still explain. A slightly better score that nobody can justify to a customer is a poor trade in a regulated business.

Build a governed data layer first

Most failed AI projects in banks fail on data, not on algorithms. I would link core banking, loan origination, payment switches, CRM, contact-centre records and app events into one controlled layer. Bureau data and trusted fraud signals are added where permitted. Every data item is tagged with its source, freshness, sensitivity, purpose, access rights and retention period. Customer identifiers are standardised, duplicates resolved and timestamps aligned, and personal identifiers are masked or tokenised in development and test environments.

One discipline deserves special mention. Loan and fraud models must be trained only on information that would have been available at the moment of the original decision. If a feature quietly includes something learned later, such as a repayment that happened after

approval, the model looks excellent in testing and disappoints in production. The data team should log the as-of date for every feature.

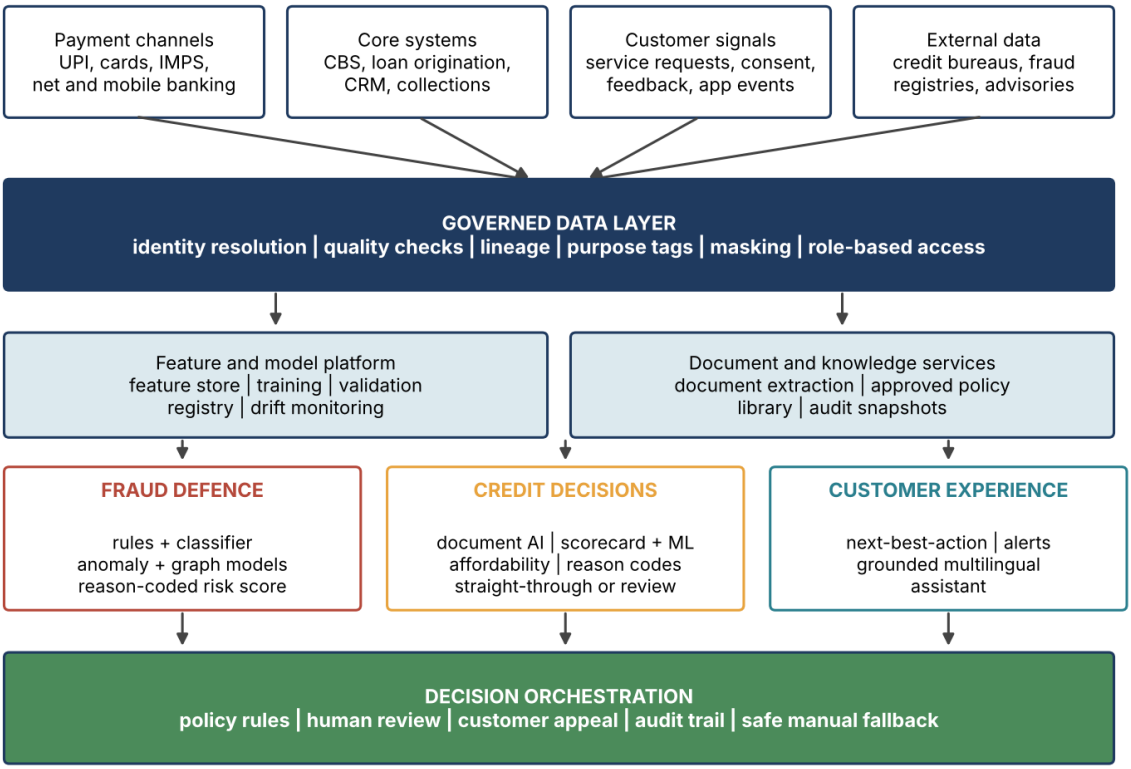


Figure 1. Proposed AI operating architecture: one governed data layer and shared platform supporting three purpose-built use cases, with a common decision and audit layer.

The shared platform provides secure data access, a feature store, independent validation, a model registry, controlled deployment, drift monitoring and audit logs. Each model gets a named business owner, a stated purpose, known limitations, a fallback procedure and a retirement date. If a model or data feed fails, the bank returns to rules or manual processing. I would deliberately start with one fraud journey and one high-volume retail loan journey. The components can be reused later, but a fraud model should not be assumed to work for credit or marketing.

3. Priority one: stop fraud earlier without annoying genuine customers

I rank fraud first because the loss is immediate and often unrecoverable, and because it erodes trust faster than a slow loan does. Fraud also has the clearest feedback loop: investigations produce labels that make the next model better. The RBI's Master Directions on fraud risk management expect banks to have early-warning capability and governance around it, and the RBI has also encouraged work on tools for detecting mule accounts, so this is an area where supervisory expectations and business need point the same way [2].

Proposed solution: a layered fraud score

I propose a near-real-time scoring service for digital payments, logins and account changes. The signals include amount and frequency, time of day, new device or SIM change, login behaviour, payee history, location mismatch, account age and patterns from confirmed cases. Four detectors work side by side:

- **Policy rules and limits** remain in place for known patterns and mandatory controls.
- **A supervised classifier** learns from labelled genuine and fraudulent events and catches combinations a human would not write as a rule.
- **Anomaly detection** compares each event with that customer's own normal behaviour, which helps with new scam types.
- **A graph model** looks at shared devices, shared beneficiaries and rapid pass-through of funds, which is how mule networks are often exposed.

The service returns a risk score with reason codes, not a hidden yes or no. A decision layer then applies proportionate action: allow low-risk activity, ask for step-up authentication when uncertain, and hold or escalate only where risk and policy justify it. High-risk cases reach a trained analyst with a short explanation of the strongest signals. A customer who is wrongly blocked needs a quick way to fix it, such as an in-app confirmation or a call-back.

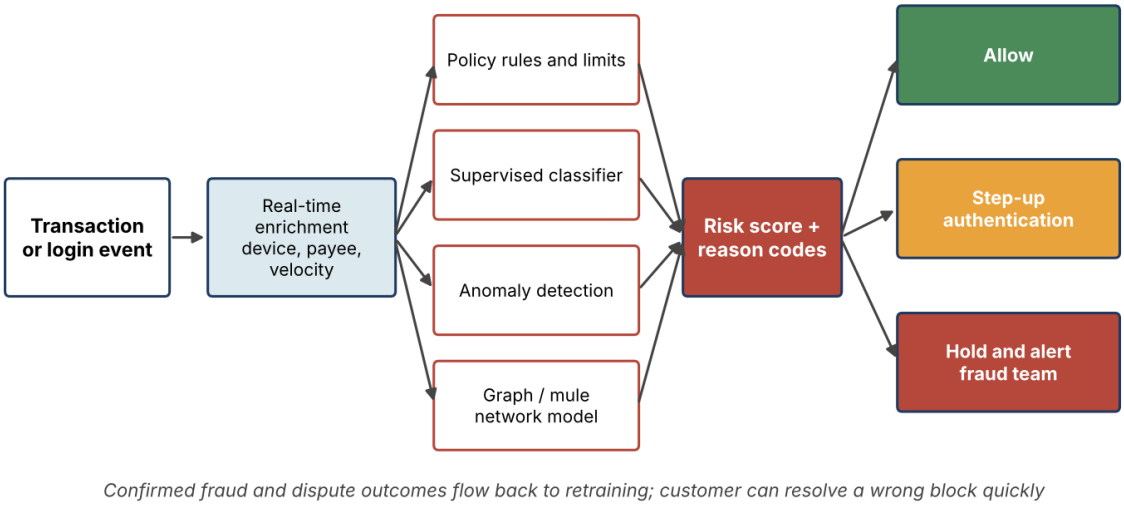


Figure 2. Real-time fraud decision flow. Four detectors feed one risk score; the action is proportionate, and confirmed outcomes flow back into retraining.

Balance detection against customer friction

Fraud is rare compared with genuine activity, so overall accuracy is a misleading measure. A model that approves everything is almost always right and catches nothing. A model that flags everything catches all fraud and stops every customer. The bank should therefore track precision (how many alerts are truly useful), recall (how many frauds are caught), false-decline rate, money lost, and time from suspicious activity to action. Thresholds should differ by channel and amount: a low-value UPI collect request and a high-value first-time international card payment do not deserve the same setting.

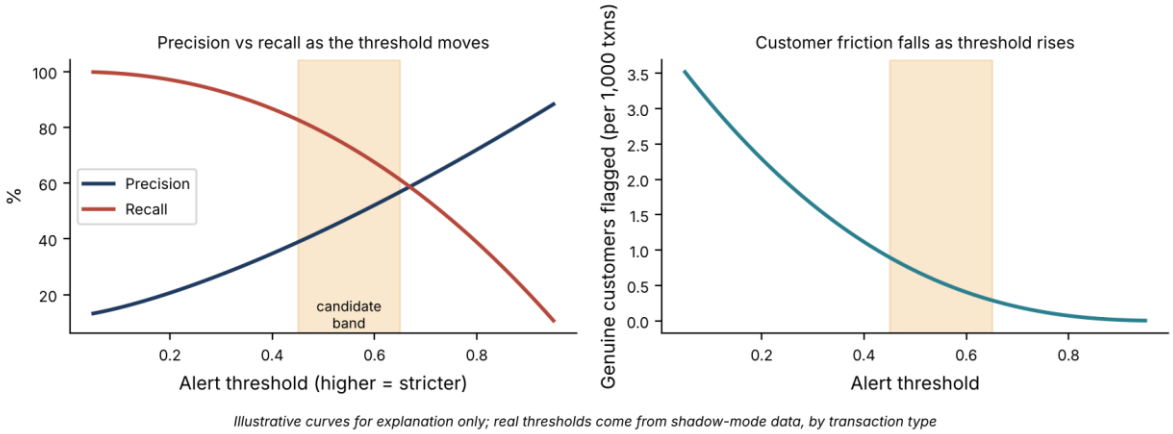


Figure 3. How the alert threshold trades precision against recall and customer friction. The curves are illustrative, not measured bank data.

Controls and the learning loop

The pilot starts in shadow mode: the model scores transactions without changing the customer journey, predictions are compared with confirmed outcomes, and thresholds are tuned before any live action. After go-live the team monitors score drift, new scam patterns, missing data, alert volumes and performance by customer segment. Existing rules, transaction limits, authentication controls and incident-response procedures stay active as a fallback. External advisories and trusted fraud signals are considered as part of the bank's own fraud-risk process, not left to the model alone. Success means fewer fraud losses and faster intervention without a matching rise in false declines, complaints or avoidable support calls.

4. Priority two: faster credit decisions that remain fair

In lending, most of the delay is administrative rather than analytical. Documents arrive in different formats, staff re-type the same data, details are checked more than once, and every file waits in the same queue regardless of how clear it is. AI can remove much of that friction. The aim is straight-through processing for clear, low-risk applications and better-supported judgement for the rest, not automatic approval for everyone.

Proposed solution: a tiered credit workflow

The workflow begins with a digital application and pulls permitted data such as bureau history, existing obligations and, where the customer agrees, account-aggregator data on income and cash flow. Document AI extracts fields from salary slips, bank statements and tax records. Extracted values are always checked against source records. If a field is missing, conflicting or low-confidence, the system asks the customer for clarification or sends the file to an officer instead of accepting the value as fact.

A supervised model then estimates the probability of default or assigns a risk band, set against a transparent scorecard as a benchmark. A separate component estimates affordability. Explicit policy rules handle eligibility, product limits and regulatory requirements, so the model never overrides a hard rule. A complex model is kept only if independent testing shows a meaningful improvement over the scorecard and its output can be explained to the bank's own risk team.

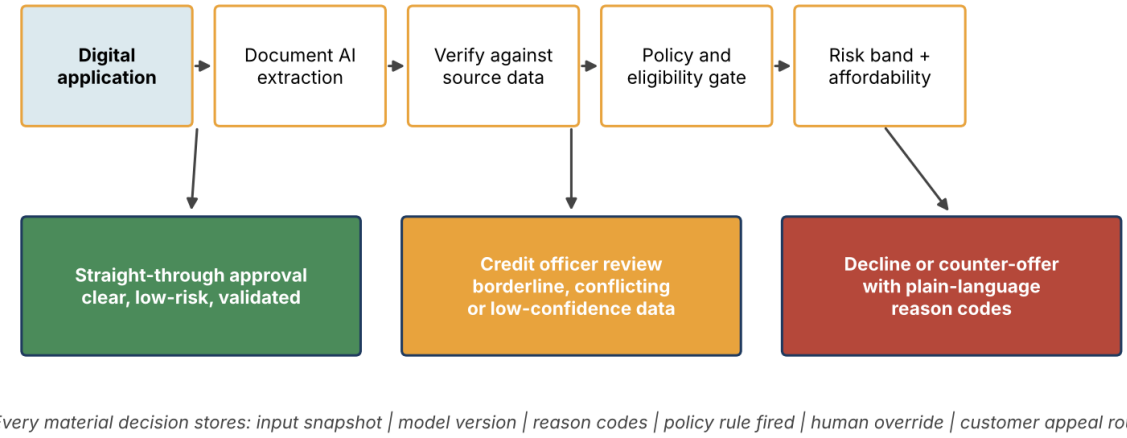


Figure 4. Tiered credit decision flow. Clear cases go straight through; uncertain cases go to a credit officer; declines carry plain-language reasons.

Make speed compatible with fairness

The model should be trained on accurate and representative data and tested across customer groups, regions, income ranges, product types and new-to-credit applicants. Protected or sensitive attributes should be excluded from decision inputs unless law permits or requires their use. Even then, variables such as pin code, occupation or transaction pattern can act as proxies, so the bank should test for them using controlled audit data. A model that looks fair on average can still treat one group worse.

For every material decision the system stores the input snapshot, model version, reason codes, policy rule applied and any human override. Borderline, inconsistent or out-of-scope applications go to a credit officer. Customers should be told what is missing, what happens next, and how to query or correct inaccurate information. A model score is not an explanation. Reason codes must make sense to the loan officer and, in plain language, to the applicant.

Pilot and validation

Begin with one high-volume, relatively standard retail product, such as a salaried personal loan or a loan against a consumer purchase. Compare the AI-supported workflow with the current process using a time-based holdout sample and a controlled pilot. Validate discrimination, calibration, stability, approval quality, early delinquency and fairness before raising the share of automated decisions. Track median and 90th-percentile turnaround alongside early repayment behaviour. Cutting average time while credit quality worsens would not count as success. Before production, the bank should review the RBI's rules on digital lending, KYC, outsourcing and data protection, since several apply directly to how lending apps and partners operate [4].

5. Priority three: personalised banking that customers actually value

Personalisation has a poor reputation because many banks equate it with more messages. I would define it differently: remove effort for the customer first, and promote products only when they are genuinely suitable. That is better for the customer and, over time, better for engagement numbers too.

A reliable customer view and useful recommendations

From permitted data on account activity, product holdings, service requests, stated preferences and interaction history, the bank builds a controlled customer view. Segmentation reveals broad needs, and a recommendation model ranks possible actions. Examples include a reminder before a recurring bill falls due, a budgeting tool for a customer who keeps asking about spending, a deposit-maturity alert, or a prompt to set up a card limit before travel. Each recommendation passes eligibility rules, suitability checks and the customer's communication preferences. A predicted click is never enough on its own.

A guarded multilingual assistant

India's customers speak many languages, and a lot of service friction is simply language friction. A conversational assistant that handles questions on charges, account features, application status and card controls in Hindi, regional languages and English can lift self-service completion and relieve call-centre load. It should use retrieval from the bank's approved and current documents so answers rest on real policy, show the source where helpful, say plainly when it does not know, and hand over to a person for anything sensitive, disputed or complex.

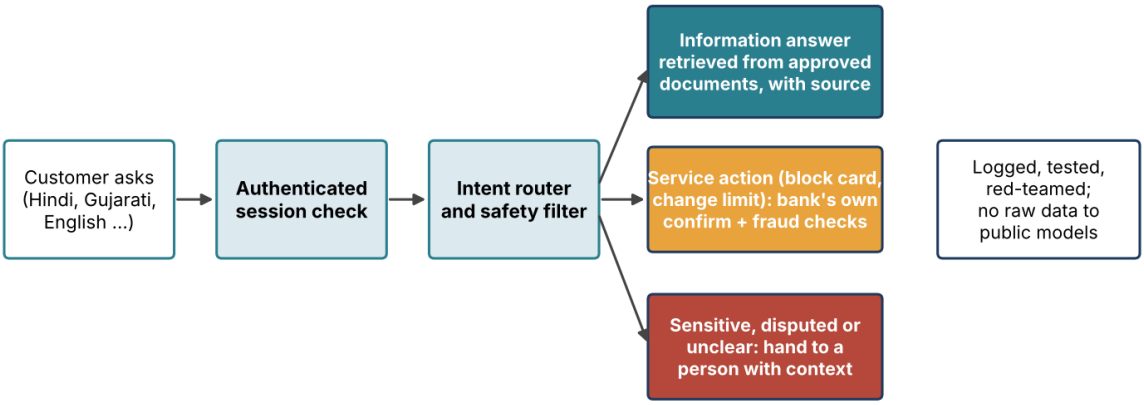


Figure 5. Guardrails around the service assistant: authenticated session, intent routing, grounded answers, controlled actions and human handover.

The assistant must be clearly presented as an AI tool and work inside the customer's authenticated session. Read-only information is kept separate from actions such as changing a limit, blocking a card or moving money, and every action uses the bank's existing authorisation, confirmation and fraud checks. Raw customer records, credentials, full card numbers and confidential documents must never be sent to a public model. The bank should also test the assistant against attempts to trick it, and review language quality with native speakers, since a translation that is fluent but wrong about a fee is worse than no answer.

Measure value, not clicks

Recommendations and assistant journeys should be tested in controlled experiments. The measures that matter are self-service completion, time to resolution, repeat contact, satisfaction, complaints, opt-outs and whether customers finish the task they started. Click-through alone can reward noisy or manipulative messages. Use frequency caps, honour communication preferences, give easy opt-outs, and review outcomes for customers who may be financially vulnerable. The FREE-AI report stresses consumer awareness, people-first

outcomes and accountability, and I would treat those as design requirements for this workstream [1].

6. Regulatory compliance, privacy and responsible AI

I would handle responsible AI as part of operational risk management, built into product design and approval from day one, not added after launch. The FREE-AI report sets out principles and strategic pillars for responsible AI in finance. It is a good framework for policy and controls, but it does not replace binding RBI directions or applicable law [1].

How each requirement is built into the design

Area	Requirement or risk	How the strategy responds
Fraud-risk governance	RBI Master Directions on fraud risk management: early warning, accountability, reporting [2]	Fraud score and alert queue feed the existing fraud-risk framework; model decisions logged; analysts keep authority to override
IT governance and resilience	RBI Master Direction on IT governance, risk, controls and assurance; outsourcing and third-party risk [3]	Model inventory, change control, tested fallbacks, vendor exit plans, audit rights over any third-party model or cloud service
Digital lending	RBI Digital Lending Directions, 2025: disclosures, data handling, no dark patterns [4]	Clear offer and reason communication; consent-based data collection; customer route to query or correct decisions
Personal data	Digital Personal Data Protection Act 2023 and DPDP Rules 2025, with phased commencement [5]	Purpose-tagged data, notice and consent management, minimisation, retention limits, breach response; compliance calendar tied to each effective date
Fairness and explainability	Risk of proxy discrimination; opaque decisions; FREE-AI principles [1]	Subgroup testing, reason codes, human review, appeal route, independent validation
Generative AI misuse	Wrong answers, data leakage, prompt attacks	Grounded answers, no raw data to public models, red-teaming, safe fallback to staff

Governance and accountability

The Board, or its designated committee, should approve the bank's AI risk appetite and policy. A cross-functional group drawn from business, risk, compliance, legal, data, technology, cybersecurity, model validation and customer protection should keep a register of all models and material third-party AI services, classify each use case by impact, and apply the heaviest review to fraud interventions and credit decisions. Every model needs two owners: a business owner who is accountable for outcomes, and a technical owner who understands how it works.

Before release, the team documents purpose, training data, limitations, test results, thresholds, human oversight, change controls and fallback arrangements. Independent validation should challenge the assumptions, not repeat the developer's tests. After launch the bank monitors drift, error rates, fairness, security incidents and complaints, and material incidents follow a documented escalation and reporting route.

Privacy and data security

For every data field the bank should be able to say why it is needed, which notice or consent applies, who can see it, how long it is kept and when it is deleted. The DPDP Act and Rules have phased commencement provisions, so the legal team should map each duty to its effective date, and the bank must keep complying with the sector-specific laws and directions that already apply to each activity. Technical measures include minimisation, encryption in transit and at rest, strong key management, least-privilege access, masking, secure development practices and tested incident response. The app should not collect contacts, gallery data or location history just because the operating system allows it. Novel or high-impact uses need a privacy impact assessment, and before using an external model or cloud service the bank should review cross-border processing, subcontractors and audit rights. Outsourcing a tool does not outsource the bank's responsibility to its customers.

Customer recourse

Customers should know when they are dealing with an AI assistant and how to reach a person. They should be able to correct wrong data and ask for review of an adverse decision. The bank also keeps a human override and a working manual fallback for outages or unexpected model behaviour. The governing principle is simple: the bank stays accountable for decisions made with AI, even when a vendor supplied the tool.

7. Implementation roadmap and measurement

An 18-month phased programme

In months 0 to 3 the bank confirms executive sponsorship, sets up AI governance, inventories existing rules and models, maps data flows, defines privacy and security controls, and selects a small set of measurable use cases. It also records baselines for fraud loss, false positives, loan turnaround, approval quality, self-service completion and complaints. In months 3 to 6 the fraud model runs in shadow mode and the credit pilot starts on one standard product. The team validates data quality and reason codes, tests stability, and trains the staff who will handle alerts and exceptions.

In months 6 to 12, validated pilots move to controlled production and a limited multilingual assistant launches using approved content. Personalisation begins only after consent flows, suitability rules, frequency limits and opt-out journeys have been tested. Months 12 to 18 focus on scaling the proven uses, independent audit, remediation of weak segments, vendor resilience checks, and retiring models that have not shown value.

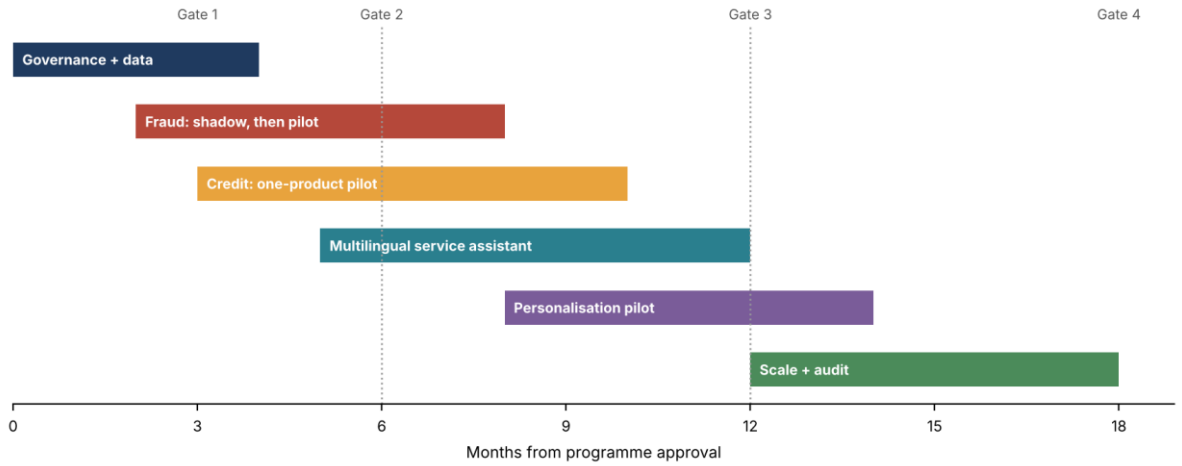


Figure 6. Indicative 18-month roadmap with four review gates. Timings are planning windows that depend on data readiness, procurement and regulatory review.

Balanced scorecard

Results should be judged across four dimensions, not on one headline number. Targets are set only after the baseline is measured, since the case gives none.

Dimension	Measures to track	Guardrail
Risk (fraud)	Confirmed fraud loss; time from first suspicious event to action; precision and recall by channel; mule accounts identified	False-decline rate and fraud-related complaints must not rise
Speed (credit)	Median and 90th-percentile turnaround; share of straight-through cases; manual rework per file	Early delinquency, override rate and fairness metrics must not worsen
Customer value	Self-service completion; first-contact resolution; satisfaction; opt-out and complaint rates; repeat contacts	Opt-outs and vulnerable-customer complaints reviewed monthly
Responsible operation	Model inventory coverage; validation completion; drift alerts; privacy or security incidents; time to remediate	No high-impact model in production without independent validation

Main risks and how to manage them

- **Poor data quality.** Treated as a gating item in months 0 to 3; no model goes live on data that fails agreed quality checks.
- **Over-blocking genuine customers.** Shadow mode, tuned thresholds, step-up authentication before hard blocks, and a fast recovery route.
- **Biased or unexplainable credit decisions.** Subgroup testing, reason codes, officer review of borderline cases, and an appeal route.
- **Vendor dependence.** Exit plans, audit rights, and a manual or rules-based fallback for each use case.
- **Staff resistance.** Fraud analysts and credit officers co-design alert screens and reason codes; their feedback becomes training data.
- **Fast-changing rules.** A compliance calendar for RBI and DPDP milestones, reviewed by the governance group each quarter.

8. Conclusion

My main conclusion is that AI can help this bank prevent fraud, decide suitable loans faster and make everyday banking less frustrating, but the gains will not come from algorithms alone. They depend on reliable data, clear ownership, careful testing, secure integration and easy access to a human. I recommend focused pilots first, then wider use only where results are measurable, explainable and controlled. In banking, trust is what makes digital change last, and each design choice in this strategy is meant to protect it.

References

[1] **Reserve Bank of India. (13 August 2025).** FREE-AI Committee Report: Framework for Responsible and Ethical Enablement of Artificial Intelligence. https://www.rbi.org.in/scripts/PublicationReportDetails.aspx?ID=1306

[2] **Reserve Bank of India. (15 July 2024).** Master Directions on Fraud Risk Management in Commercial Banks (including Regional Rural Banks) and All India Financial Institutions. https://www.rbi.org.in/Scripts/BS_PressReleaseDisplay.aspx?prid=58294

[3] **Reserve Bank of India. (7 November 2023; effective 1 April 2024).** Master Direction on Information Technology Governance, Risk, Controls and Assurance Practices. https://www.rbi.org.in/Scripts/BS_ViewMasDirections.aspx?id=12562

[4] **Reserve Bank of India. (2025).** Reserve Bank of India (Digital Lending) Directions, 2025. See RBI Annual Report 2024-25. https://www.rbi.org.in/scripts/AnnualReportPublications.aspx?Id=1436

[5] **Ministry of Electronics and Information Technology, Government of India. (2023, 2025).** Digital Personal Data Protection Act, 2023, and Digital Personal Data Protection Rules, 2025. https://www.meity.gov.in/static/uploads/2024/02/Digital-Personal-Data-Protection-Act-2023.pdf

Note: the figures in this report are conceptual designs and illustrative curves, not data from a real bank. Regulatory references should be re-checked against the latest RBI and MeitY publications before implementation.